

Deterministic Quantum Statistics from Real-Number Quantum Theory and Null-Shell Causality

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Abstract

This paper presents a deterministic reconstruction of quantum statistical behavior based on a real-number formulation of quantum theory combined with time-symmetric null-shell causality. Real-number quantum theory (RNQT) demonstrates that the complex structure of Hilbert space is representational rather than ontological: all operational predictions of quantum mechanics can be recovered using real vector spaces alone. Null-shell causality treats physical interactions as completed two-boundary events occurring at lightlike separation, with retarded and advanced contributions participating symmetrically. We show that these two frameworks fit naturally together: the two real components underlying a complex amplitude correspond physically to the retarded and advanced legs of a null-shell transaction. Once amplitude squaring is assumed, the geometry of these contributions yields the Born probability law without adjustable parameters. Deterministic single-event outcomes arise by partitioning continuous coupling through a symmetry-derived threshold, while ensemble statistics reproduce standard quantum predictions. Bell correlations emerge from boundary-conditioned phase selection that relaxes measurement independence without violating locality or no-signaling. The resulting framework provides a deterministic account of quantum probabilities grounded in null geometry and real-number Hilbert-space structure.

Keywords

quantum foundations; real-number quantum theory; null-shell causality; Born rule; Bell correlations; retrocausality; direct-action electrodynamics; determinism

1 Introduction

Quantum mechanics is widely regarded as intrinsically probabilistic. While this view is operationally successful, it leaves unresolved whether quantum probabilities reflect fundamental indeterminism or instead arise from incomplete access to underlying physical structure. Einstein rejected the former possibility, insisting that physical probabilities should ultimately admit a deterministic explanation.

Two independent developments substantially alter the foundations of this debate. The first is *real-number quantum theory* (RNQT), which demonstrates that the full operational content of quantum mechanics can be reconstructed using real Hilbert spaces without assuming complex numbers as fundamental entities [1]. In this formulation, the imaginary unit functions as an internal rotation acting on pairs of real components rather than representing an ontological degree of freedom.

The second development is a class of time-symmetric, direct-action approaches to interaction, here referred to as *null-shell causality*, inspired by Fokker’s invariant action and Wheeler–Feynman absorber theory [2, 3]. In this framework, observable events correspond to completed two-boundary transactions between emission and absorption, constrained to lightlike separation and involving both retarded and advanced contributions.

These two developments are rarely considered together. In this paper we argue that they are naturally complementary. The two real components underlying a complex amplitude in RNQT correspond physically to the retarded and advanced legs of a null-shell interaction. Quantum probabilities then emerge geometrically from the overlap of these contributions, rather than from an irreducible stochastic postulate.

2 Real-Number Quantum Theory

RNQT reconstructs quantum mechanics using real Hilbert spaces by embedding complex numbers into 2×2 real matrices that represent internal rotations [1]. The imaginary unit i is represented by

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \tag{1}$$

acting on real vectors $(a, b)^T$ corresponding to the complex amplitude $a + ib$.

Standard quantum operations—density matrices, observables, tensor products, and measurement statistics—are faithfully reproduced by restricting dynamics to a symmetric subspace of real matrices closed under a modified tensor product. All operational predictions remain unchanged. Complex structure is therefore representational rather than ontologi-

cal, encoding relations between real components rather than introducing additional physical degrees of freedom.

3 Null-Shell Causality

Null-shell causality treats interactions as lightlike contacts between worldlines. Observable events correspond to closed retarded–advanced loops connecting emission and absorption. This modernizes direct-action electrodynamics, in which fields are not independent entities but arise from boundary-matching conditions [2, 3].

Let λ_e and λ_a denote emission and absorption phases relative to a polarizer axis θ . The retarded and advanced contributions are

$$R = \cos(\lambda_e - \theta), \quad A = \cos(\theta - \lambda_a). \quad (2)$$

The transaction amplitude is

$$\mathcal{A} = RA = \cos(\lambda_e - \theta) \cos(\theta - \lambda_a), \quad (3)$$

with associated weight

$$W = |\mathcal{A}|^2. \quad (4)$$

These two real factors play the role of the two components otherwise packaged into a complex amplitude.

4 Emergence of Born Statistics

Assuming absorber phases uniformly distributed on $[0, 2\pi)$, the detection probability becomes

$$P(\theta|\lambda_e) \propto \int_0^{2\pi} \cos^2(\lambda_e - \theta) \cos^2(\theta - \lambda_a) d\lambda_a. \quad (5)$$

Normalization yields

$$P(\theta|\lambda_e) = \cos^2(\lambda_e - \theta), \quad (6)$$

recovering Malus’s law and the Born rule. The result follows directly from null geometry once amplitude squaring is assumed.

5 Deterministic Threshold

Continuous coupling may be discretized into deterministic outcomes by introducing a symmetry threshold

$$\text{Detect if } |\mathcal{A}| > T_{\text{crit}}. \quad (7)$$

A natural choice is $T_{\text{crit}} = 0.5$, which symmetrically partitions alignment space and reproduces Born statistics under averaging. Individual outcomes are deterministic functions of $(\lambda_e, \theta, \lambda_a)$, while ensemble behavior remains probabilistic.

6 Bell Correlations

For a polarization singlet with phases $(\lambda, \lambda + \pi/2)$, local amplitudes at settings a, b are

$$\mathcal{A}_A = \cos(\lambda - a), \quad (8)$$

$$\mathcal{A}_B = \sin(\lambda - b). \quad (9)$$

The hidden-variable density

$$\rho(\lambda|a, b) \propto \cos^2(\lambda - a) \sin^2(\lambda - b) \quad (10)$$

yields the correlation

$$E(a, b) = -\cos(2(a - b)), \quad (11)$$

matching quantum predictions and maximal CHSH violation. Measurement independence is relaxed through boundary conditioning, while locality and no-signaling are preserved.

7 No Dice, Niels

Einstein’s rejection of fundamental indeterminism is often summarized by his remark that “God does not play dice.” In the present framework this statement is interpreted not as a rejection of quantum statistics, but as a challenge to the assumption that probabilistic outcomes lack an underlying causal explanation.

Here, standard quantum statistics are reproduced while assigning definite outcomes to individual events determined by boundary conditions. Apparent randomness arises from ignorance of absorber microstates rather than intrinsic indeterminism. The “dice” of quantum mechanics are statistical shadows cast by incomplete boundary information on null shells.

8 Scope and Non-Claims

This framework does not permit superluminal signaling or controllable retrocausation. It does not predict deviations from standard quantum mechanics under ordinary conditions, and it does not reintroduce Bell-local hidden variables. The approach offers a deterministic reconstruction that is operationally equivalent to standard quantum theory while providing a distinct causal interpretation.

9 Relation to Other Retrocausal Models

The framework shares time symmetry with the transactional interpretation and two-state-vector formalism, and is closely related to boundary-based approaches developed by Price and Wharton [4, 5]. Its distinguishing feature is the explicit identification of real amplitude components with null-shell geometry via RNQT.

10 Testability

Anisotropic absorber environments or suppression of advanced contributions may lead to small deviations from ideal statistics while preserving no-signaling. Experiments with engineered absorbers or asymmetric cavities could therefore probe the framework.

11 Conclusions

RNQT removes complex amplitudes from ontology, while null-shell causality provides a physical interpretation of the resulting real structure. Born statistics emerge geometrically, deterministic outcomes arise from symmetry thresholds, and Bell correlations follow from boundary conditioning. Quantum randomness is thus epistemic rather than fundamental.

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Conflict of Interest

The author declares no conflict of interest.

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